

Long Context vs. RAG: Strategies for Processing Long Documents in LLMs

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Abstract

Large Language Models (LLMs) excel at zero- and few-shot learning but are restricted by the length of context windows when processing long documents. Two strategies have emerged to overcome this limitation: (1) Long Context (LC) methods, which extend or compress transformer architectures to input more text; and (2) Retrieval-Augmented Generation (RAG), which integrates external knowledge sources via embedding- or index-based retrieval. This half-day tutorial offers a unified, beginner-friendly introduction to both approaches. We first review transformer fundamentals—positional encoding, attention complexity, and common LC techniques. Next, we explain the classic RAG pipeline and recent RAG strategies, alongside evaluation metrics and benchmarks. We also analyze recent empirical studies to highlight strengths, limitations, and trade-offs of LC vs. RAG in terms of scalability, computational cost, and retrieval effectiveness. We conclude with best practices for real-world deployments, emerging hybrid architectures, and open research directions, equipping IR researchers and practitioners with actionable guidelines for processing long documents in LLMs.

CCS Concepts

• **Information systems** → **Language models; Retrieval effectiveness; Question answering.**

Keywords

Retrieval Augmented Generation, Long Context Understanding, Large Language Model

ACM Reference Format:

Xinze Li, Yushi Bai, Bowen Jin, Fengbin Zhu, Liangming Pan, and Yixin Cao. 2025. Long Context vs. RAG: Strategies for Processing Long Documents in LLMs. In *Proceedings of the 48th International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR '25)*, July 13–18, 2025, Padua, Italy. ACM, New York, NY, USA, 4 pages. <https://doi.org/10.1145/3726302.3731690>



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ACM ISBN 978-1-4503-XXXX-X/2018/06
<https://doi.org/10.1145/3726302.3731690>

1 Motivation

Large Language Models (LLMs) [5] have demonstrated strong zero- or few-shot capabilities in open-ended question answering [32]. However, they face challenges such as hallucinations [14, 25], lacking real-time information and domain-specific knowledge [26, 35], among others. A common solution is to enhance LLMs with external memory to provide reliable and up-to-date data sources. Yet, incorporating additional content is constrained by the limited context window of LLMs. In real-world applications such as legal documents and medical report understanding, LLMs must handle queries that require external knowledge from very long documents beyond their context window.

To address this, two main approaches are adopted: (i) building models with long context windows to read in more information (**LC**) [7, 10, 30], and (ii) employing retrievers to include text segments relevant to the query (**RAG**) [2, 11, 15]. Understanding the differences between these approaches is vital for researchers and practitioners, as each comes with unique strengths, limitations, and implementation challenges. A key challenge is that simply increasing model size is not a feasible way to encode all world knowledge, and naively feeding very long inputs into standard models is computationally expensive and potentially ineffective. On the other hand, integrating an external knowledge source via retrieval raises its own issues: How do we ensure the model gets most relevant information but not just semantically similar ones?

Therefore, a "battle" has developed within the community: some argue that extending the context window may eventually replace retrieval mechanisms, while others maintain that RAG remains essential [33] for filtering noise and ensuring precise, up-to-date information. Although several surveys [16, 18, 31] have compared these approaches, they often arrive at divergent conclusions. Since 2023 there has been research devoted to handling longer inputs and combining retrieval with LLM. However, these papers arrive at divergent conclusions. This discrepancy raises important questions: What experimental choices or definitions of "long context" lead to these different outcomes? How do we fairly evaluate the trade-offs between model complexity, computational cost, and retrieval effectiveness?

In this tutorial, we offer a beginner-friendly introduction to the two strategies for processing long documents. We start by covering the basics of LLMs and the transformer architecture, then explain

how long-context techniques allow models to input more text, while how retrieval techniques search for more relevant information to aid LLM generation. By reviewing historical developments of techniques and benchmarks from both approaches, we will explore this ongoing "battle" between LC and RAG, discussing key challenges and insights. This tutorial aims to equip audiences interested in IR with better understanding on the strategy selection when processing long document for LLMs.

2 Objectives

The primary goal of this half-day tutorial is to provide attendees with a clear and foundational understanding of how LLMs can be enhanced with extended context and retrieval-based techniques. By the end of the tutorial, attendees will:

Understand LLM Context Handling: Comprehend how modern LLMs process input text and what limitations arise with long inputs. We will explain in simple terms the architecture of transformers and why standard models have fixed context lengths (e.g. positional encoding [6] and attention complexity [29]). Attendees will learn the classic methods that long-context models overcome these limits – including 1. Sparse and Local Attention [8] 2. Low-rank and Approximations [12] 3. Recurrent Mechanisms[9] 4. Compressed Memory Modules [22].

Learn Retrieval-Augmented Generation Basics: Learn the core concepts of RAG [19], where an external knowledge corpus is used to provide additional information to the model. We will cover the classic technical workflow of RAG: how a user query is transformed into embedding, how relevant documents are retrieved based on similarity match, and how the retrieved content is integrated into the LLM’s input to produce a more informed response. We will introduce the 3 groups of retrieval technique, chunk based [13, 23], index based [20], and summarization based [24] retrieval. Attendees will learn their similarities in core concepts and how their difference results in different retrieval performance.

Compare LC and RAG – Strengths and Weaknesses: Develop an ability to contrast the LC and RAG approaches in various dimensions. We will present a comparative analysis highlighting scenarios where long context models excel versus where retrieval augmentation performs better. Attendees will see a breakdown of trade-offs – for instance, LC can use original context order and full passages, whereas RAG can pick out the most relevant snippets from a large knowledge base but might lose some surrounding context. We will also discuss limitations: LC models face computational and memory challenges as context grows, while RAG systems depend on retrieval quality and can be brittle if the search fails to find key information.

Gain practical insights and Identify Future Directions: Equip attendees with actionable guidelines for implementing LC and RAG in real-world systems. We will provide concrete recommendations, such as prioritizing LC for tasks requiring narrative coherence and RAG for dynamic knowledge integration. For future directions, we will discuss emerging trends like new architectures, and more efficient retrieval methods. Attendees will have a clear understanding for advancing research in context-aware LLMs, balancing performance and scalability.

Table 1: Schedule of the Presentation

Time	Session
0:00 - 0:20	Session 1: Introduction
0:20 - 1:00	Session 2: Retrieval-augmented Generation
1:00 - 1:30	Session 3: Long Context LLMs
1:30 - 2:00	Coffee break
2:00 - 3:00	Session 4: RAG v.s. LC Comparison
3:00 - 3:20	Session 5: Summary and Future Directions
3:20 - 3:30	Session 6: Q&A

3 Relevance

This tutorial sits at the intersection of Information Retrieval (IR) [21] and Natural Language Processing, making it highly relevant to the SIGIR community. Retrieval-augmented LLMs are essentially the combination of search engines with text generation models: the retrieval component leverages classic IR principles (indexing, ranking, query expansion) to provide content to the generative model. We will explicitly connect tutorial topics to IR research – for example, discussing retrieval techniques used to fetch documents in RAG systems. Attendees from the IR field will recognize how their expertise in ranking algorithms and efficient search can enhance LLM performance, and conversely, NLP researchers will see how LLMs can be applied to improve search and recommendation results.

The topic of “Long Context vs. RAG” is timely and cutting-edge, touching on numerous works across SIGIR, ACL, EMNLP, ICLR, and Neurips conferences. Attendees will be pointed to relevant research papers from both the IR community (e.g., on efficient indexing and document retrieval) and the NLP community (on large context language modeling and knowledge-enhanced generation) to show the broad applicability. This collaboration also significant to many recent advances in conversational search and knowledge-grounded AI assistants, making our tutorial highly relevant for SIGIR 2025 participants interested in the future of information access and recommendation.

To our knowledge, the most related tutorial is the ACL 2023 Tutorial on Retrieval-based Language Models and Applications [1], which teaches on by then up-to-date progress of architectures and learning algorithms of retrieval-based LMs for diverse NLP tasks. Our tutorial focuses on the comparison of extending context window length (LC) and retrieval-augmented generation (RAG) approaches for long document handling of LLMs, including techniques, benchmarks, applications and experiment designs on various domains. Another relevant tutorial is from SIGIR 2024 Tutorial on Large Language Model Powered Agents for Information Retrieval [34], which explores emerging techniques for deploying LLM-powered agents across search engines, social networks, recommender systems, and conversational platforms. They focus on using LLM as a search engine, while our tutorial focuses on whether using long context LLM to take in the long documents as a whole can replace traditional retrieval.

4 Format and Schedule

This half-day tutorial consists of 6 sections, including a Q&A section. The planned schedule is as shown in Table 1:

4.1 Introduction (20 Minutes)

Kick off the tutorial with an engaging overview of the sessions. Briefly introduce the key concepts of long-context models and retrieval-augmented generation (RAG). Highlight the challenges LLMs face with limited context and the opportunities provided by both extended context and retrieval techniques. Explain that there has been a battle between the two techniques in handling long documents. We also introduce the core concepts and basic technical details of large language models, including the transformer architecture, the basics of self-attention mechanism [28], and the auto-regressive models' token generation. This section is a preliminary and is very basic for most attendees. However, since our target audiences are beginners, we make sure all attendees are equipped with these knowledge so we can proceed to explaining the techniques to extend context window length.

4.2 Retrieval-Augmented Generation (40 Minutes)

We introduce the RAG paradigm that provide relevant context to LLMs. We explain how RAG integrates external knowledge by retrieving relevant text segments from a larger corpus, which are then incorporated into the generation process. We also cover the retrieval pipeline, including the use of embedding-based searches and traditional Information Retrieval techniques, and discuss the benefits (such as up-to-date information and reduced memory load) along with potential challenges. We explain a history of the retrieval techniques and what are nowadays effective and popular. We also provide an overview of benchmark datasets used to evaluate long-RAG systems, ranging from open-domain QA [17] to fact verification [27]. Attendees will learn about common evaluation metrics (e.g., ROUGE, EM/F1, perplexity, and retrieval recall)

4.3 Long Context LLMs (30 Minutes)

We present an in-depth explanation on the methods that extend the context length of LLMs, we discuss the classic strategies including 1) **Efficient Transformer Architectures**: Sparse attention, low-rank approximations. 2) **Memory-Augmented Approaches**: Recurrence and compressive memory modules. 3) **Hierarchical and Chunking Methods**: Breaking long texts into chunks and using summarization or sequential processing. This section explains how each technique helps overcome the fixed context window limitations, and what are their advantages and disadvantages respectively. We elaborate on the capabilities of Long Context LLMs including context length and preferred task. We also provide an overview of benchmark datasets [3, 4] used to evaluate long-context LLMs, ranging from long-document summarization to reading comprehension.

4.4 RAG v.s. LC Comparison (60 Minutes)

In this session, we explore the ongoing "battle" between long-context (LC) approaches and retrieval-augmented generation (RAG) methods. Some researchers argue that by extending the context window, LLMs can eventually replace retrieval mechanisms, while others maintain that RAG remains essential for filtering noise and providing up-to-date, precise information. Despite several surveys comparing these strategies, divergent conclusions persist. During

this hour-long discussion, we will review and analyze approximately 10 recent research papers that either compare or combine LC and RAG approaches. We will:

- Present key empirical findings and case studies to highlight the varied perspectives in the literature.
- Explore the experimental choices and definitions of "long context" that contribute to differing outcomes.
- Evaluate the trade-offs between model complexity, computational cost, and retrieval effectiveness.

By examining these dimensions, we aim to equip attendees with a comprehensive understanding of the strengths and limitations of each approach, while also identifying avenues for further improvement in tackling long-document retrieval challenges.

4.5 Summary and Future Directions (20 Minutes)

We recap the key takeaways from the tutorial, and summarize the main points regarding the technical approaches, empirical findings, and practical applications of LC and RAG. We conclude the tutorial with an overview on the open research questions, including new architectures, and more efficient retrieval methods.

4.6 Q&A (10 Minutes)

We open the floor to the attendees for an interactive Q&A session. This segment will allow participants to ask questions, share insights, and discuss practical applications.

5 Materials

To maximize the tutorial's value, we will provide comprehensive materials and resources to all participants, including slide deck, codes, and dataset ¹:

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¹The codes and dataset are available at https://github.com/lixinze777/LC_VS_RAG

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